

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Semester Examination – Summer 2022

Course: B. Tech.

Branch : CE / CSE / CS

Semester : VII

Subject Code & Name: Blockchain Technology [BTCOE704A]

Max Marks: 60

Date: 24/08/2022

Duration: 3.45 Hrs.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

[12]

- | | |
|---|------------|
| A) What is the definition of Blockchain? Describe the characteristics of blockchain technology. | Understand |
| B) Define the Merkle Tress and discuss the working of Merkle Trees. Also, Explain the benefits of Merkle tree in blockchain technology. | Understand |
| C) Explain in details the various usage of cryptography preferred in the blockchain. | Understand |

Q.2 Attempt the following questions.

[12]

- | | |
|---|------------|
| A) Describe in brief the block structure of bitcoin blockchain. | Understand |
| B) Differentiate between Proof of Work (PoW) and Proof of Stake (PoS) protocol in blockchain. | Analyze |

Q. 3 Solve Any Two of the following.

[12]

- | | |
|--|------------|
| A) Discuss the several challenges faced while designing an application using permissioned blockchain frameworks. | Understand |
| B) Explain in detail working of the Practical Byzantine Fault Tolerance System. | Understand |
| C) Enlist the several design goals to choose the right blockchain consensus protocol. | Remember |

Q.4 Solve the following questions.

[12]

- | | |
|---|------------|
| A) Explain any two blockchain-based applications can be preferred in the domain of finance for trading. | Understand |
| B) Explain any two blockchain-based applications can be preferred in the domain of supply chain management. | Understand |

Q. 5 Solve Any Two of the following.

[12]

- | | |
|--|------------|
| A) Differentiate between Permissioned and Permissionless Blockchain. | Analyze |
| B) Enlist and explain the major components of Hyperledger Fabric. | Understand |
| C) Describe the core components of Ripple ecosystem. | Understand |

*** End ***